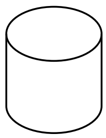


RDB

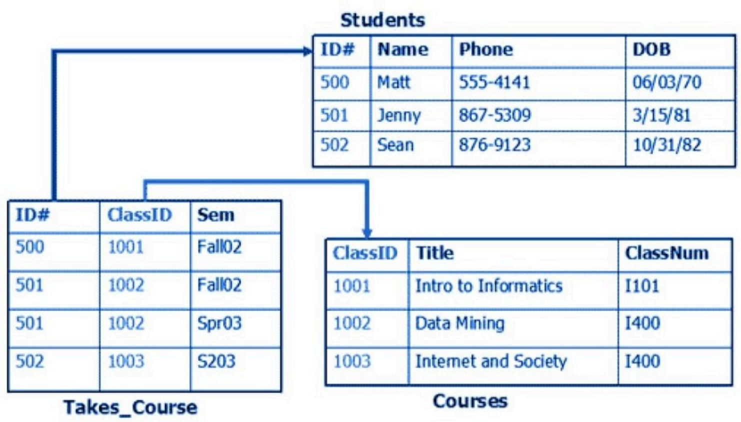
- RDBMS: Relational Database Management System
- RDB: Relational Database
- Standardd relational databases run on single computer / server
- Example of Relational Databases: MySQL, Oracle Database, Postgres Database

Data in RDB's are stored in one or more tables. Each tables contain records. It also stores relationships among recrods of one table to the records of another table.

For example, if Facebook stored their social media data in a RDB, they may store user's personal information in a table called users, all user posts in a table called posts, and store all comments in another table.

Similar type of data is stored in a single table to aid in querying, grouping and analyzing. A facebook user will have a single record on the users table. If the user created posts, their corresponding post records will be in the posts table. The same table will contain post records of other users. These records will have information about the corresponding user id to signify the relationship among the user and their posts.

**Examples of Relational Data:**



Students

| ID# | Name | Phone | DOB |
|---|---|---|---|
| 500 | Matt | 555-4141 | 06/03/70 |
| 501 | Jenny | 867-5309 | 3/15/81 |
| 502 | Sean | 876-9123 | 10/31/82 |

| ID# | ClassID | Sem |
|---|---|---|
| 500 | 1001 | Fall02 |
| 501 | 1002 | Fall02 |
| 501 | 1002 | Spr03 |
| 502 | 1003 | S203 |

Takes_Course

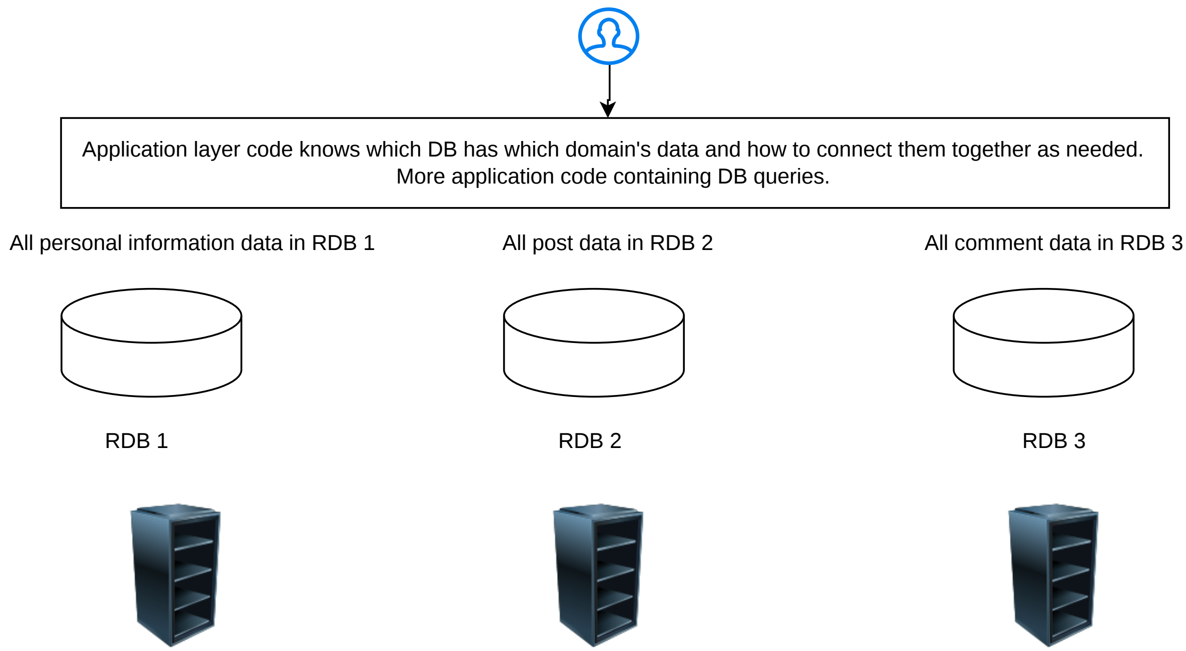
| ClassID | Title | ClassNum |
|---|---|---|
| 1001 | Intro to Informatics | I101 |
| 1002 | Data Mining | I400 |
| 1003 | Internet and Society | I400 |

Courses

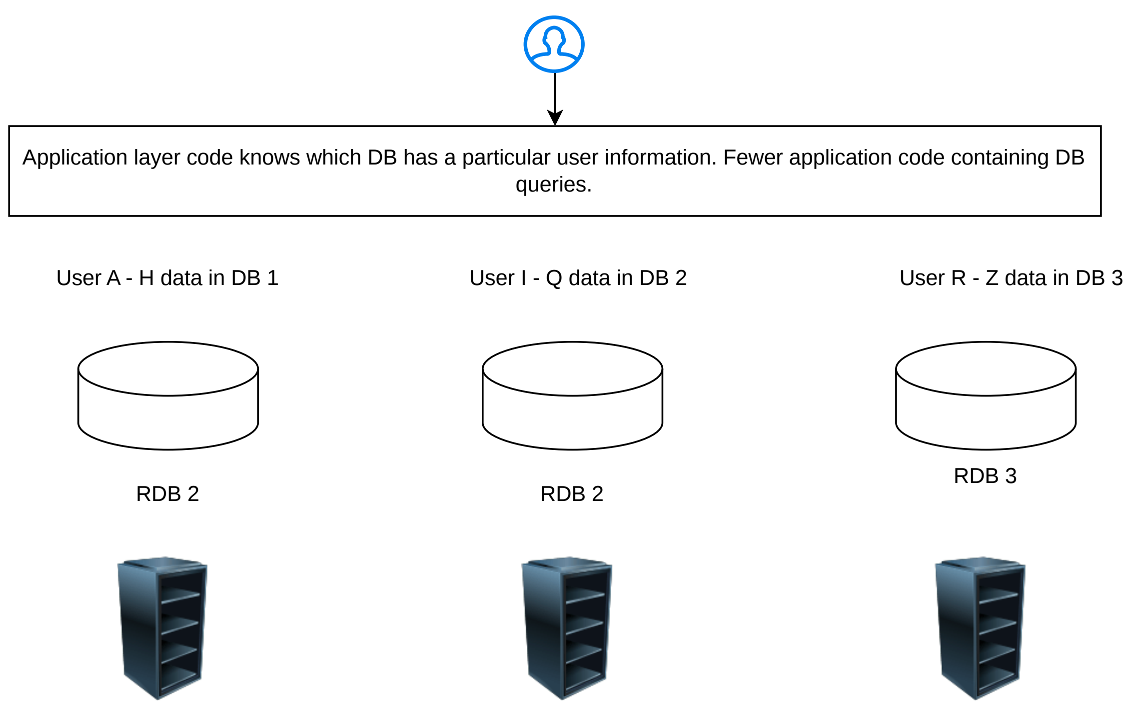
Server

Why is there a limitation in different databases in very large scale?
- these DBs are run and built for a single computer/server system (in contrast to a distributed system)
- there's physical limitation in building the highest performing single server system :
  - Compute (CPU)
  - Storage (DISC)
  - Memory (RAM)

**Sharding / Partitioning** by **Domain** to accomodate a lot of data in less expensive servers:




**Sharding / Partitioning** by **Range** to accomodate a lot of data in less expensive servers:

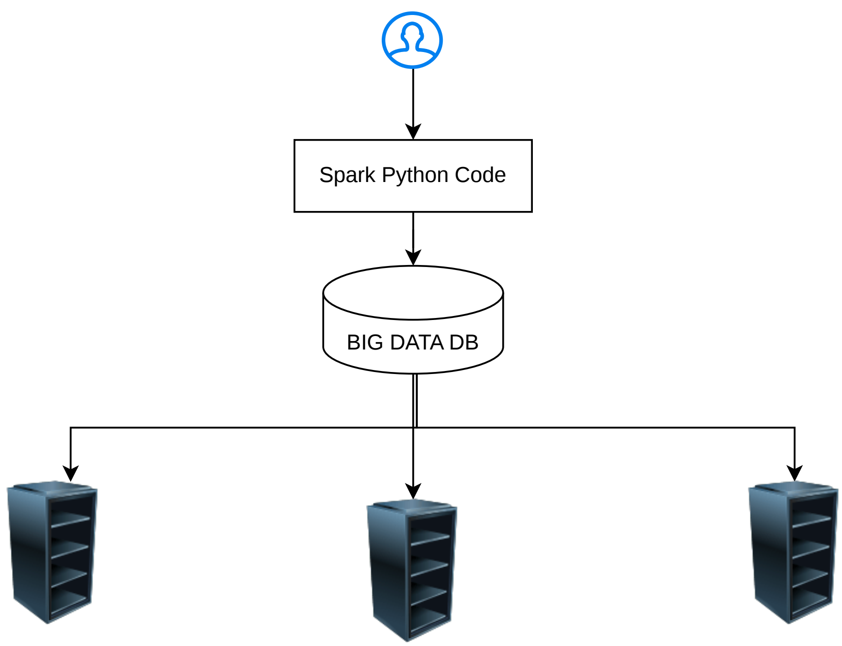


OLTP: Online Transaction Processing
OLAP: Online Analytical Processing

RDBMSs are catered toward OLTP.
Big Data systems are *mostly* catered toward OLAP solution.

Big Data:

- We have a lot of unstructured data to analyze
- We want to store them in the disk to reduce cost rather than storing in a database intantace
- We want to store data that doesn't fit in single relational database or it will is difficult/not worth it to partition or shard data in a easy, elegant way




Big Data System (overly simplified)